

Git Commands

- **Git:** Global Information Tracker
- **Branches:**
 - `master` is the default branch.
 - `git branch` - View all branches.
 - `git branch <branch-name>` - Create a new branch.
 - `git checkout <branch-name>` - Switch to a branch.
 - `git merge <branch-name>` - Merge a branch into the current branch.
 - `git branch -d <branch-name>` - Delete a branch.
 - `git branch -D <branch-name>` - Forcefully delete a branch.
 - `git branch -r` - View all remote branches.
- **Staging and Committing:**
 - `git add <filename>` - Stage changes for commit.
 - `git add .` - Stage all changes in the directory.
 - `git commit -m "msg"` - Commit staged changes with a message.
 - `git commit --amend -m "new message"` - Modify the latest commit message.
- **Repository Management:**
 - `git init` - Initialize a new empty repository.
 - `git clone <url>` - Clone a repository.
 - `git pull` - Fetch and merge changes from the remote repository.
 - `git push` - Push local changes to the remote repository.
 - `git push origin <branch-name>` - Push to a specific branch in the remote repository.
- **Viewing Changes:**
 - `git status` - Show the status of the working directory and staging area.
 - `git log` - View commit history.
 - `git log --oneline` - View commit history in a single line format.
 - `git show <commit-id>` - Show details of a specific commit.
- **Configuration:**
 - `git config --global user.name "Jithu"` - Set the global username.
 - `git config --global user.email "jithendarramagiri@gmail.com"` - Set the global email.
 - `git config --list` - List all configuration settings.
- **Advanced Features:**
 - `git stash` - Save changes temporarily.
 - `git stash list` - List all stashed changes.
 - `git stash pop` - Apply the most recent stashed changes and remove them from stash.
 - `git cherry-pick <commit-id>` - Apply changes from a specific commit.
- **Tagging:**
 - `git tag <tag-name>` - Create a new tag.
 - `git checkout <tag-name>` - Checkout a specific tag.
 - `git tag -d <tag-name>` - Delete a tag.
- **Resetting Changes:**
 - `git reset HEAD <filename>` - Unstage a file.

- `git reset --soft <commit-id>` - Move the HEAD pointer to a previous commit without changing the working directory.
- `git reset --mixed <commit-id>` - Move the HEAD pointer and unstage files.

Linux Commands

- **Basic Navigation and File Management:**
 - `pwd` - Print the current working directory.
 - `ls` - List files in the current directory.
 - `ls -la` - List all files, including hidden ones.
 - `mkdir <directory-name>` - Create a new directory.
 - `touch <filename>` - Create a new empty file.
 - `rm -rf <filename|directory-name>` - Forcefully remove a file or directory.
 - `cd <directory-name>` - Change to another directory.
 - `cd ..` - Move up one directory level.
 - `clear` - Clear the terminal screen.
 - `cat <filename>` - Display the contents of a file.
- **User Management:**
 - `whoami` - Show the current user.
 - `adduser <username>` - Add a new user.
 - `passwd <username>` - Change a user's password.
 - `userdel -r <username>` - Delete a user and their home directory.
 - `usermod -c "comment" <username>` - Modify a user's information.
- **System Information:**
 - `hostname` - Display the server name.
 - `hostnamectl set-hostname <new-name>` - Change the server hostname.
 - `df -h` - Display disk space usage.
 - `free -g` - Show memory usage in GB.
 - `top` - Display the task manager.
- **File Operations:**
 - `cp <source> <destination>` - Copy files.
 - `mv <source> <destination>` - Move or rename files.
 - `tar -cvf <archive.tar> <directory>` - Create a tar archive.
 - `tar -xvf <archive.tar>` - Extract a tar archive.
 - `zip <archive.zip> <files>` - Create a zip archive.
 - `unzip <archive.zip>` - Extract a zip archive.
- **Searching and Filtering:**
 - `grep <pattern> <file>` - Search for a pattern in a file.
 - `find / -name <filename>` - Search for a file by name.
 - `diff <file1> <file2>` - Compare two files.
 - `wc <file>` - Count lines, words, and characters in a file.
 - `head -n <number> <file>` - Show the first n lines of a file.
 - `tail -n <number> <file>` - Show the last n lines of a file.
- **Editing Files:**

- `vim <filename>` - Open a file in Vim editor.
- `:i` - Switch to insert mode in Vim.
- `esc` - Switch to command mode in Vim.
- `:wq` - Save and exit Vim.
- `dd` - Delete a line in Vim.
- `yy` - Copy a line in Vim.
- `p` - Paste copied content in Vim.